

Mohammed Hassan

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Profile

Motivated third-year Computer Science student specializing in cybersecurity, with hands-on experience in full-stack development and software testing within government and enterprise environments. Strong foundation in Python, JavaScript/TypeScript, SQL, and modern frameworks, with experience building scalable search systems and document pipelines. Analytical problem-solver with an agile mindset, strong attention to detail, and a passion for secure, efficient systems. Proven ability to collaborate in technical teams and adapt quickly to new technologies.

Education

Bachelor of Science in Computer Science, with an option in Cyber Security (Co-op)

Apr. 2027 (Expected)

Carleton University, Ottawa, ON

GPA: 3.6 / 4

Project Experience

Radiance Vault | *Python, Meilisearch, PyWebView, InstantSearch.js, Tailwind CSS*

- Developed a secure document search and classification platform for CBSA as part of a small dev team, enabling Border Services Officers to label, index, and retrieve documents; implemented a microservices-inspired architecture with real-time search, automated indexing, secure desktop UI, and document parsing pipelines (PDF, DOCX, MHTML, images), optimizing query performance to sub-50ms.

ObjectInsight | *JavaScript, ML5.js, HTML5, CSS3, Materialize CSS*

- Developed a browser-based real-time object detection application using the device camera; implemented live video inference with ML5.js, adjustable frame rates, responsive canvas resolution, and an intuitive UI allowing users to toggle AI detection and visualize detected objects in real time.

Hospital Record Management System | *C++*

- Developed a console-based hospital record management system using C++, implementing file-based persistent storage with binary I/O to add, search, display, modify, delete, and transpose patient records; applied structured programming, class-based design, conditional logic, and manual data handling to manage patient demographics through a menu-driven interface.

Work Experience

Junior Developer

Apr. 2025 – Present

Canada Border Services Agency (CBSA), Ottawa, ON

- Collaborated in an agile team to design and implement Radiance Vault, a full-stack document management and search system.
- Developed backend services in Python (Typer, Starlette, Shiny) and built a cross-platform desktop UI using PyWebView.
- Integrated Meilisearch for real-time, typo-tolerant search and configured secure API key-based authentication.
- Designed a document ingestion pipeline (PDF, DOCX, MHTML, images) and improved scalability through process orchestration and structured logging.

Solutions Test Specialist

Sep. 2024 – Apr. 2025

March Networks, Ottawa, ON

- Designed detailed test plans and cases in JIRA and XRAY, ensuring full coverage of requirements and functionality.
- Assisted clients with network configurations, NAT traversal, and firewall troubleshooting for seamless integration.
- Developed and executed both manual and automated test cases during regression and sanity testing cycles.
- Conducted end-to-end testing for advanced video surveillance systems, including Linux/Windows DVRs, NVRs, and enterprise-grade servers.

Technical Skills

Languages: Python, JavaScript/TypeScript, C, C++, C#, Java, R, MySQL, HTML5, CSS3

Developer Tools: Git, GitHub, Linux, JIRA, XRAY, Docker, Pydantic

Framework / Libraries: React.js, Node.js, Express.js, Angular, Tailwind CSS, Shiny, PyWeb-View, Meilisearch

Leadership and Achievements

Carleton Computer Science Society (CSS)

Present

Carleton University

CU Blueprint

2023-2024

Carleton University

Three-time recipient of the Henry Marshall Tory Scholarship

2022-2025

Carleton University